

“This structure has features which are of **considerable biological interest.**”

Francis Crick and James Watson, 1953

An expert X-ray crystallographer, Franklin had produced a clear image of DNA—Photograph 51—showing its double helix structure. Franklin was working alongside Maurice Wilkins, but their relationship was hostile, and Wilkins showed Photograph 51 to Crick and Watson without her knowledge. Watson and Crick had already experimented with helical structures but without success. Franklin's image helped them confirm the double helix and complete their model.

Double helix

The double helix structure of DNA resembles a twisted ladder, with the phosphoric acid and sugar strands acting as the ladder's uprights and the paired “bases” forming the rungs. What was crucial was the pairing of the bases—Erwin Chargaff, an Austrian scientist, had previously noted that the A and T bases always appeared in equal ratios, as did the G and C bases. Watson and Crick realized that each rung of the ladder must be

composed of two bases (either A and T or C and G) bonded together and that the order of the bases formed the code for genetic information. They deduced that when the ladder splits in half—this is often described as the molecule “unzipping”—each “half ladder” bonds with newly formed DNA components to form two new genetically identical double helices.

Scientific breakthrough

In March 1953, Crick and Watson completed the double helix model of DNA, and they published their discovery in a landmark paper in April. Franklin's contribution to their work received only passing acknowledgment. Considered one of the greatest scientific discoveries of the 20th century, Crick and Watson's work marked a turning point in our understanding of genetics and paved the way for huge advances in molecular biology. In 1962, along with Maurice Wilkins (Franklin having died in 1958), they were awarded the Nobel Prize in Physiology or Medicine.

Crick (right) and Watson made their famous 3D model out of various materials, including paper and metal plates. ►

